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Waraporn Phathaisuang, Chanoknan Boonlupyanan, Parid Vetchaiyo, Phatthranit Laocharoen, Peeraphan Limritnonthiphak, Parineak Romtrairat, Thitima Jintanawan & Gridsada Phanomchoeng

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# An Accessible 2D Video-Based Gait Analysis Framework with Model-Based Illustration of Joint Torque and Power Trends

Waraporn Phathaisuang<sup>1</sup>, Chanoknan Boonlupyanan<sup>1</sup>, Parid Vetchaiyo<sup>1</sup>, Phatthranit Laocharoen<sup>1</sup>, Peeraphan Limritnonthiphak<sup>1</sup>, Parineak Romtrairat<sup>1</sup>, Thitima Jintanawan<sup>1,\*</sup>, and Gridsada Phanomchoeng<sup>1,2,\*</sup>

<sup>1</sup>Department of Mechanical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok 10330, Thailand

<sup>2</sup>Center of excellence for human-robot collaboration and systems integration, Faculty of Engineering, Chulalongkorn University, Bangkok 10330, Thailand.

\*Corresponding authors: Thitima Jintanawan (thitima.j@chula.ac.th), and Gridsada Phanomchoeng (gridsada.phanomchoeng@gmail.com).

## Abstract

This paper presents an accessible single-camera gait analysis framework that uses a standard RGB video camera and open-source software to extract sagittal-plane joint kinematics and illustrate model-based kinetic behavior. The proposed data pipeline converts RGB video frames into 2D joint trajectories using marker-based tracking, which are subsequently processed to compute joint angles and angular velocities. These kinematic variables are then integrated into a planar three-link inverse-dynamics model implemented in MATLAB® Simscape Multibody™, enabling the estimation of joint torque and power profiles from visually captured motion.

The modeling implementation is first evaluated using a published reference dataset to verify the consistency of the inverse-dynamics computation, yielding normalized RMS differences below approximately 2% for joint torque profiles and below 5% for joint power profiles. Camera-derived joint trajectories are then applied to generate qualitative torque and power waveforms across the gait cycle. Geometric consistency analysis indicates stable reconstruction of proximal segment lengths, while larger variability is observed in distal segments due to projection effects and depth limitations inherent to monocular 2D tracking.

The extracted joint-angle and angular-velocity profiles preserve characteristic waveform structures commonly observed in sagittal-plane gait analysis. Although torque and power magnitudes differ from reference data due to planar kinematic inputs and simplified ground-reaction-force assumptions, their temporal patterns remain qualitatively consistent across the gait cycle. The framework is intended for trend-level kinetic illustration, exploratory motion analysis, and instructional use in environments where laboratory-based gait analysis systems are unavailable.

**Keywords** Gait analysis, Inverse dynamics, Biomechanics, Camera, Gait assessment, Joint angle

## Introduction

Gait analysis—the systematic study of human walking patterns—is widely used in clinical assessment and biomechanical research. Even subtle gait deviations can indicate neurological or musculoskeletal disorders and are commonly used to support diagnosis and rehabilitation monitoring [1]–[4]. Clinically, gait measurements assist physiotherapists and clinicians in evaluating treatment outcomes [1], [5]. Beyond healthcare, gait analysis also contributes to sports science, injury prevention, and the development of prosthetic or exoskeleton systems where understanding joint motion is essential for device design [1], [6]–[8]. In addition, gait mechanics are relevant to emerging technologies such as wearable energy harvesters that convert walking-induced mechanical energy into electrical power for low-power electronics [9]–[11]. These diverse applications highlight the growing need for gait analysis tools that are both informative and accessible.

Traditional quantitative gait analysis relies on laboratory systems equipped with multi-camera motion capture and force plates or pressure walkways [12], [13]. These systems provide detailed three-dimensional kinematic and kinetic measurements and enable joint moments and power to be computed through inverse dynamics using measured ground reaction forces and segment parameters [14], [15]. Optical motion capture is therefore widely regarded as a reference standard in rehabilitation research [3], [16]. However, such laboratory systems require specialized infrastructure, trained personnel, and high equipment costs, limiting their accessibility outside dedicated research facilities [3]. As a result, gait assessment in many clinical environments still relies on visual observation and subjective scoring methods that lack quantitative objectivity [3].

To improve accessibility, researchers have explored alternative approaches such as wearable sensors and video-based motion analysis [17], [18]. Wearable systems commonly employ inertial measurement units (IMUs) to measure gait dynamics and quantify spatiotemporal parameters such as stride length and gait timing [1], [20]. These sensors have also been used to monitor gait changes during neurological rehabilitation [12], [21]. However, IMU-based systems require careful sensor placement and calibration and may be affected by drift, misalignment, and soft-tissue artifacts [1].

Video-based gait analysis offers an alternative that avoids body-worn instrumentation. Marker-based approaches track anatomical landmarks using conventional cameras and computer vision algorithms to extract joint trajectories from video recordings [14], [22]. Advances in pose estimation, including deep-learning-based methods, have further enabled monocular systems to recover two-dimensional joint coordinates and, in some cases, estimate three-dimensional motion [14], [23]. Multi-camera systems can improve spatial reconstruction accuracy [14], [24]. Nevertheless, differences relative to laboratory motion capture systems are still reported, particularly in joint rotations and kinetic estimation [14], [25].

Although these accessible systems can measure spatiotemporal and kinematic parameters, estimating joint kinetics—such as forces, torques, and power—remains more challenging. In laboratory studies, inverse dynamics relies on measured ground reaction forces combined with three-dimensional kinematics [14], [15]. Joint moment and power profiles are important for interpreting gait mechanics and informing the conceptual

design of assistive or wearable devices [7], [26]. When direct force measurements are unavailable, however, kinetic estimation depends on simplified assumptions and scaled inputs, and the resulting values should be interpreted as model-based approximations rather than direct physiological measurements [1], [14].

Recent work has introduced integrated camera-based platforms for accessible gait analysis, such as OpenCap, which estimates kinematics and kinetics from multi-view smartphone recordings using musculoskeletal modeling in OpenSim. Although such systems reduce equipment requirements compared with laboratory motion capture, they still require multi-camera calibration, musculoskeletal modeling, and computationally intensive optimization frameworks.

In contrast, this study focuses on a simplified and transparent framework based on marker-based two-dimensional tracking and a planar inverse-dynamics model implemented in MATLAB® Simscape Multibody™. Rather than aiming to replicate laboratory-grade biomechanical measurements, the framework emphasizes accessibility and conceptual motion analysis.

Specifically, this study presents a 2D gait analysis pipeline that integrates standard camera video with marker-based tracking to estimate lower-limb joint angles in the sagittal plane. The inverse-dynamics model implemented in Simscape Multibody is first examined using a published reference dataset containing measured kinematics and ground reaction forces to verify the consistency of the modeling formulation. This step evaluates the correctness of the dynamic model rather than validating the camera-based measurement pipeline. After this verification step, camera-derived joint trajectories are used to explore qualitative torque and power trends under simplified modeling assumptions.

The main contributions of this work are summarized as follows:

- Accessible 2D joint kinematic extraction: Lower-limb joint angles (hip, knee, ankle) are obtained using marker-based video tracking in the sagittal plane with minimal instrumentation.
- Model-based illustration of joint torque and power trends: A planar inverse-dynamics model implemented in Simscape Multibody is used to illustrate qualitative gait-cycle torque and power behavior.
- A practical pipeline for exploratory analysis: The framework integrates video acquisition, tracking software, and dynamic modeling for instructional use and preliminary gait analysis.

Overall, this work demonstrates how accessible two-dimensional kinematic measurements can be combined with structured inverse-dynamics modeling to explore qualitative gait mechanics while explicitly acknowledging the limitations associated with simplified measurements and modeling assumptions.

The remainder of this paper is organized as follows. Section 2 describes the methodology of the proposed gait analysis framework, including the experimental setup, data acquisition, calibration procedures, kinematic computation, and the implementation of the inverse-dynamics model. Section 3 discusses the resulting kinematic and kinetic trends, highlighting key observations, modeling assumptions, and sources of discrepancy. Section 4 presents potential applications of the framework in educational, engineering,

and exploratory motion-analysis contexts. Finally, Section 5 summarizes the main findings and outlines directions for future research.

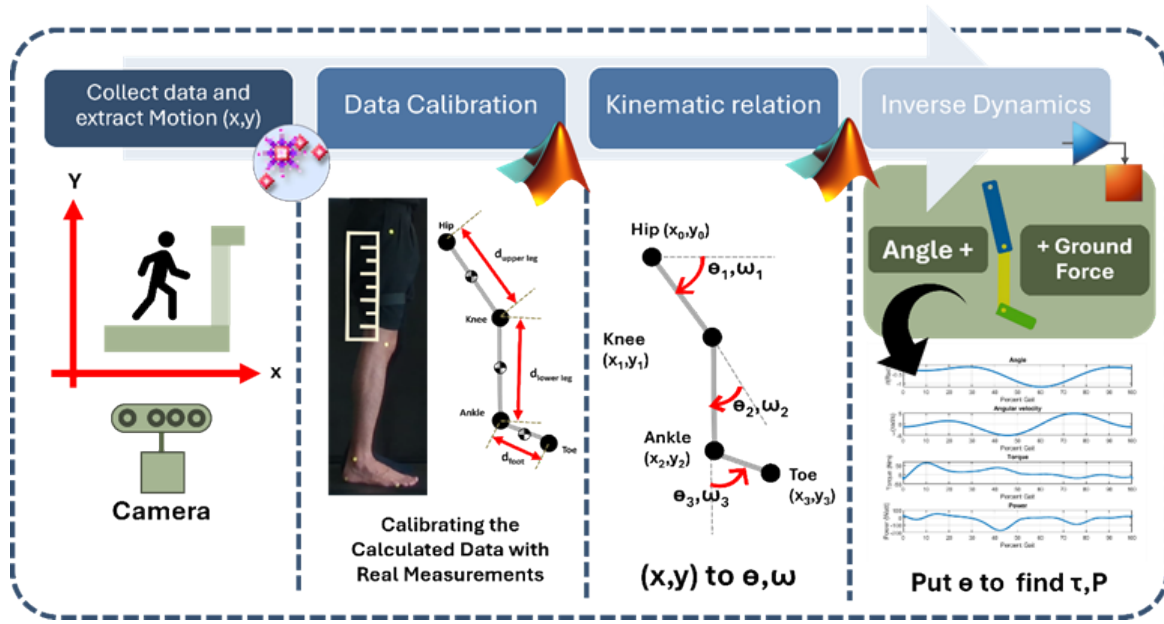
## Material and methods

### Gait Cycle Overview

A gait cycle is defined as the interval between two successive initial contacts (IC) of the same foot and is conventionally divided into stance and swing phases [12]. The stance phase begins at IC, when body weight is transferred to the supporting limb, and ends at foot-off (FO), accounting for approximately 60% of the cycle. The swing phase follows FO and continues until the next IC, representing the period of limb advancement and preparation for ground contact, typically comprising the remaining 40% of the cycle [3], [27]. Within the stance phase, periods of double support and single support occur as weight shifts between limbs. In this study, this standard gait-cycle definition was adopted for time normalization and for presenting kinematic and model-based kinetic results across participants and reference datasets.

### Overview of the Framework

The proposed gait analysis framework consists of four primary stages, as illustrated in Fig. 1. First, a video recording of the subject's walking motion is captured using a consumer-grade camera, with visual markers placed on anatomical landmarks to facilitate structured tracking. Second, the recorded video is processed using open-source tracking software to extract two-dimensional (X and Y) joint coordinates over time. Third, the coordinate data are calibrated and analyzed to estimate lower-limb joint angles—including hip ( $\theta_1$ ), knee ( $\theta_2$ ), and ankle ( $\theta_3$ )—using geometric relationships in the sagittal plane. Finally, an inverse-dynamics implementation in MATLAB® Simscape Multibody™ whose computational consistency was examined using a published reference dataset is employed to generate model-based simulations of joint torque and power trends from the extracted kinematics. The framework emphasizes accessible 2D kinematic analysis, with kinetic outputs presented as illustrative model-driven results derived under explicitly stated assumptions.



**Fig. 1.** Overview of the gait analysis framework.

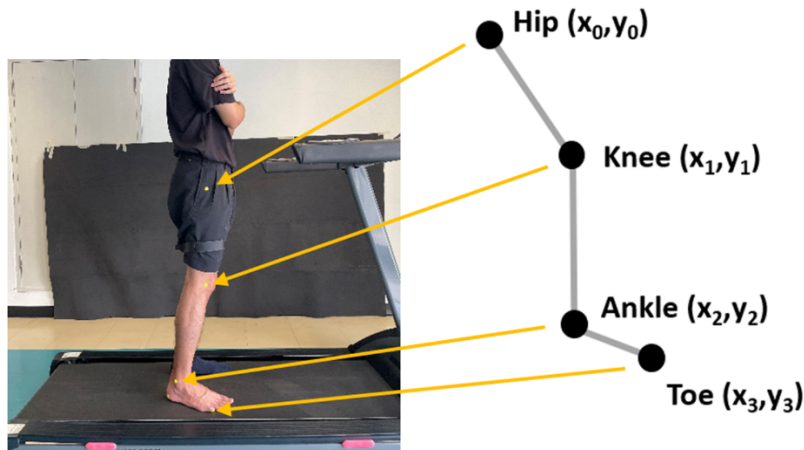
## Experimental Setup and Data Acquisition

A single healthy adult male (22 years old; height: 174 cm; mass: 59 kg) participated in this exploratory study. The analysis was conducted in the sagittal plane. Four visual markers were placed on anatomical landmarks of the right lower limb—hip, knee, ankle, and toe—corresponding to coordinates  $(x_0, y_0)$  to  $(x_3, y_3)$ , as illustrated in Fig. 2(a). Marker placement was guided by visible joint centers to reduce ambiguity in planar tracking. To minimize visual interference and marker confusion, the contralateral limb was visually masked using a dark sock.

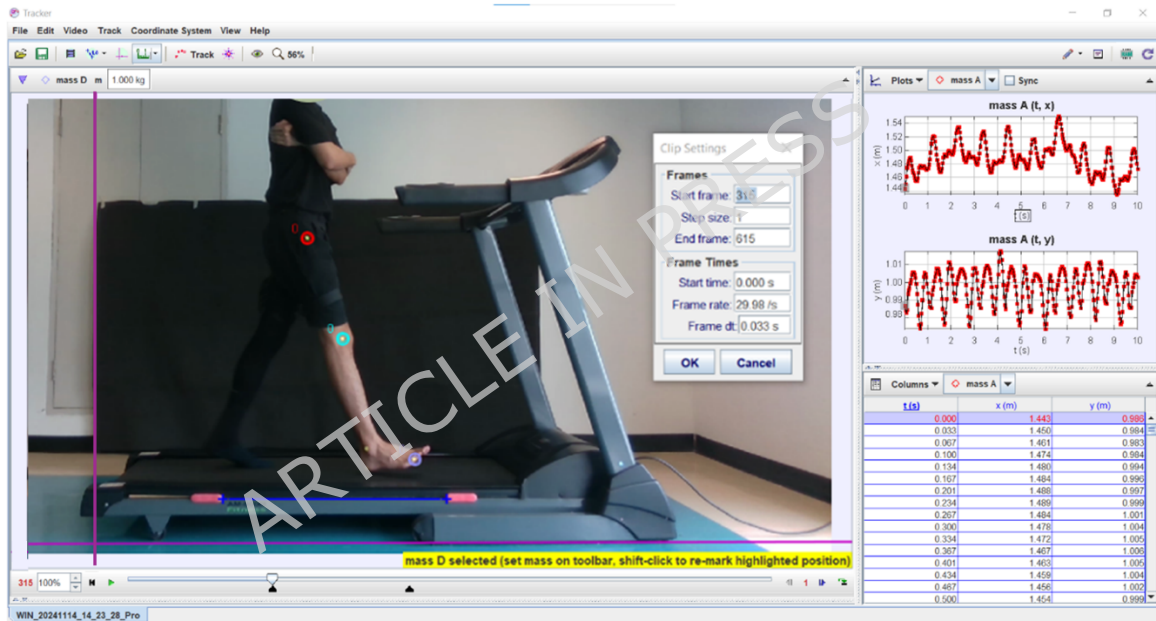
Walking was performed on a treadmill at a constant speed of 4.8 km/h under controlled indoor lighting conditions. A side-view camera was positioned perpendicular to the walking direction to approximate planar motion capture and reduce perspective distortion.

Motion was recorded using an Intel® RealSense™ D435 camera at 30 FPS in RGB mode, with depth sensing disabled to maintain consistency with 2D kinematic extraction. Video frames were processed using Tracker (Open Source Physics framework) [28], which extracted two-dimensional (x,y) marker coordinates in image space.

To focus on steady-state walking, the initial acceleration phase and final deceleration phase were excluded. Frames were selected based on visually consistent stride patterns rather than outcome-based criteria, yielding approximately 10 seconds ( $\approx 300$  frames) of continuous steady gait. Multiple consecutive gait cycles were segmented from this interval. The extracted joint-coordinate time series were smoothed using cubic spline interpolation to reduce frame-to-frame tracking noise, followed by cycle normalization and averaging to obtain representative kinematic trajectories for analysis. Fig. 2(b) presents the calibration setup and representative tracking output.



(a) Marker attachment on the participant and experimental setup.



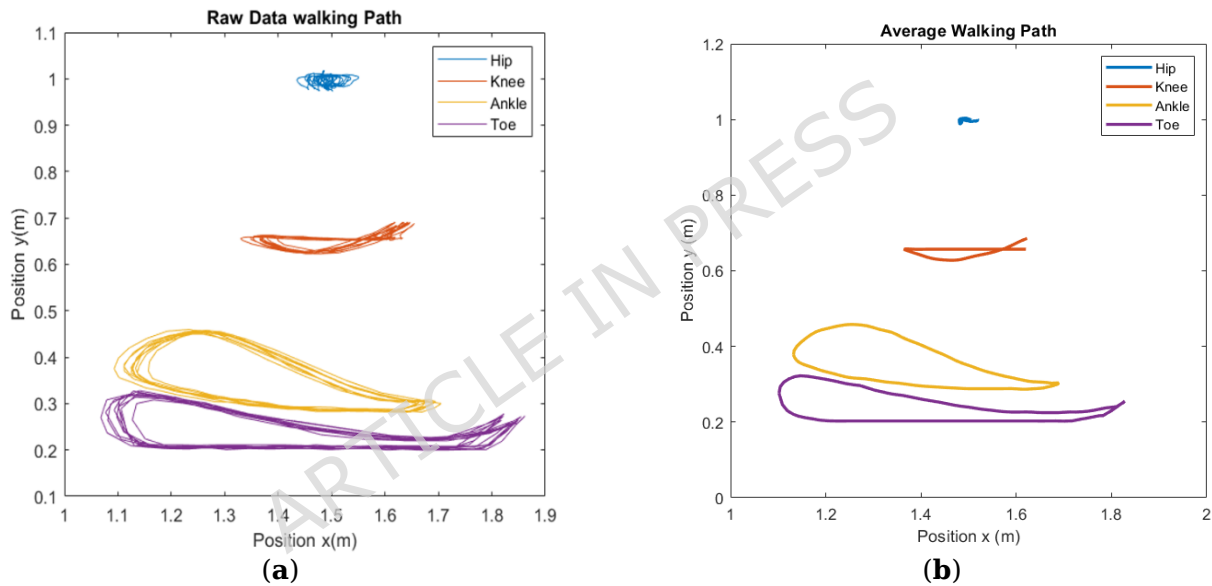
(b) Tracker program setup and axis calibration.

**Fig. 2.** Experimental setup and video tracking configuration for gait analysis.

A reference coordinate system was defined within the tracking software, and spatial calibration was performed using a known physical distance measured directly in the recording plane, such as a calibrated length marker placed within the camera field of view. This calibration procedure converted pixel coordinates into metric units (meters), enabling consistent estimation of segment lengths and joint trajectories within the sagittal plane. Because calibration was performed using a single-plane projection, the measurements remain subject to perspective and depth-related distortions inherent to monocular 2D video capture.

Following calibration, marker trajectories were extracted using the software's tracking algorithm, producing time-series (x,y) coordinates for each anatomical landmark. These coordinates served as inputs for subsequent kinematic analysis and model-based kinetic simulation. The extracted coordinates were visually inspected for continuity to minimize tracking discontinuities caused by temporary occlusion or marker blur.

The tracked trajectories were plotted to visualize cyclical gait patterns. As illustrated in Fig. 3(a), joint paths exhibited repeated closed-loop patterns characteristic of steady walking. To reduce stride-to-stride variability while preserving representative motion patterns, three consecutive steady-state gait cycles were segmented and averaged following temporal normalization, resulting in the representative trajectory shown in Fig. 3(b). The hip joint demonstrated greater variability compared to distal joints, which may be attributed to pelvis motion and out-of-plane components that cannot be fully resolved in a planar 2D side-view configuration.



**Fig. 3.** Walking trajectory obtained from marker tracking: **(a)** Raw X and Y coordinate data extracted from the video over 10 seconds. **(b)** Average X and Y coordinate trajectory.

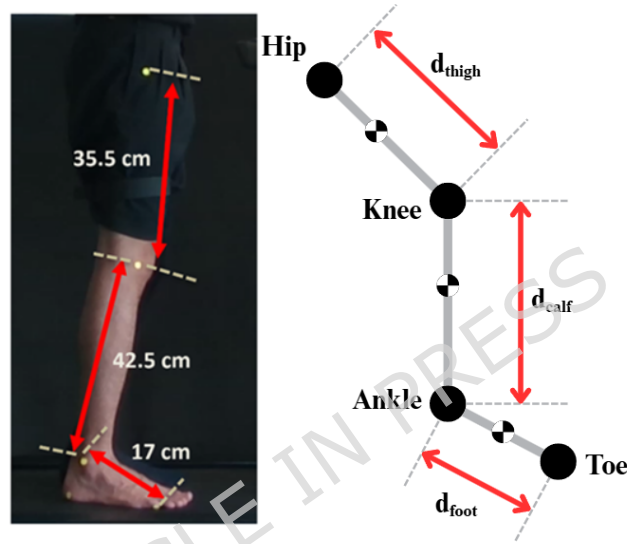
### Data Calibration and Geometric Consistency Evaluation

The tracked coordinate data were exported to MATLAB for spatial calibration and post-processing. Pixel coordinates were converted to metric units based on the previously defined planar calibration reference within the camera field of view. Because calibration was performed using a single-plane projection, the resulting measurements reflect sagittal-plane approximations and remain subject to perspective and depth-related distortion.

To evaluate geometric consistency of the tracking process, the physical lengths of three lower-limb segments—the thigh, shank, and foot—were measured directly prior to data acquisition and used as reference values. These measurements were compared with segment lengths computed from the calibrated two-dimensional joint coordinates using

Eq. (1), as illustrated in Fig. 4. This comparison provides an internal consistency check of segment reconstruction within the 2D framework, rather than a full biomechanical validation of joint kinematics.

Observed deviations between measured and reconstructed segment lengths are primarily attributed to projection effects, minor marker-placement variability, and soft-tissue movement. These deviations are expected to propagate into subsequent inverse-dynamics simulations, and therefore represent a source of uncertainty that is considered in the interpretation of model-based kinetic trends.

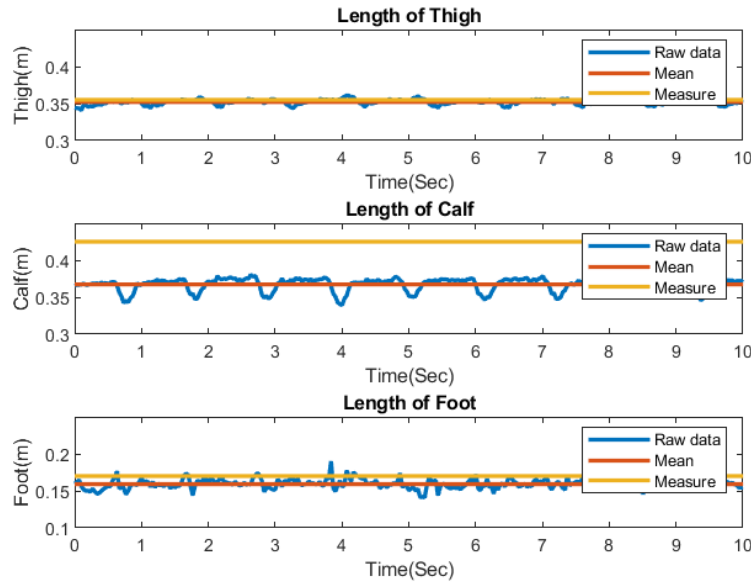


**Fig. 4.** Real-world measurements of lower-limb segments used for validating coordinate-based length calculations.

$$d = \sqrt{(x_1^2 - x_2^2) + (y_1^2 - y_2^2)} \quad (1)$$

where  $d$  represents the Euclidean distance between two joint coordinates in the sagittal plane. The computed segment lengths were compared with directly measured physical segment lengths to evaluate geometric consistency within the 2D tracking framework.

Fig. 5 presents the reconstructed lengths of the thigh, shank, and foot over a 10-second steady-state walking interval (blue curves). The temporal mean values are indicated in red, while the corresponding physical measurements are shown in yellow. All values were derived from calibrated X-Y coordinates using Eq. (1). This comparison serves as an internal consistency check of segment reconstruction rather than a comprehensive biomechanical validation.



**Fig. 5.** Calculated Segment Lengths of Thigh, Calf, and Foot During a 10-Second Gait Recording.

To quantify variability, the range of each reconstructed segment length was computed as the difference between its maximum and minimum values over time. Because true anatomical segment lengths remain constant, this temporal variation reflects tracking sensitivity to projection effects, marker placement variability, and soft-tissue motion. Additionally, the mean deviation between reconstructed and physically measured lengths was calculated to characterize systematic bias. The resulting variability and mean-difference metrics are summarized in Table 1.

The thigh segment exhibited relatively low temporal variation (range: 0.020 m, 5.63%) and minimal mean deviation (0.003 m, 0.85%), indicating stable reconstruction under sagittal-plane tracking. The shank segment demonstrated moderate variability (range: 0.041 m, 9.65%) and a larger mean deviation (0.058 m, 13.65%), likely influenced by compounded projection effects and joint center approximation. The foot segment showed the highest temporal variation (range: 0.052 m, 30.59%), which is attributable to rapid distal motion, depth ambiguity, and increased sensitivity to out-of-plane rotation in a 2D side-view recording. In addition, distal segments such as the foot exhibit larger apparent length variation because the toe marker may move relative to the ankle marker due to foot rotation during gait. However, its mean deviation remained comparatively small (0.007 m, 4.12%), suggesting limited systematic bias despite frame-to-frame fluctuation.

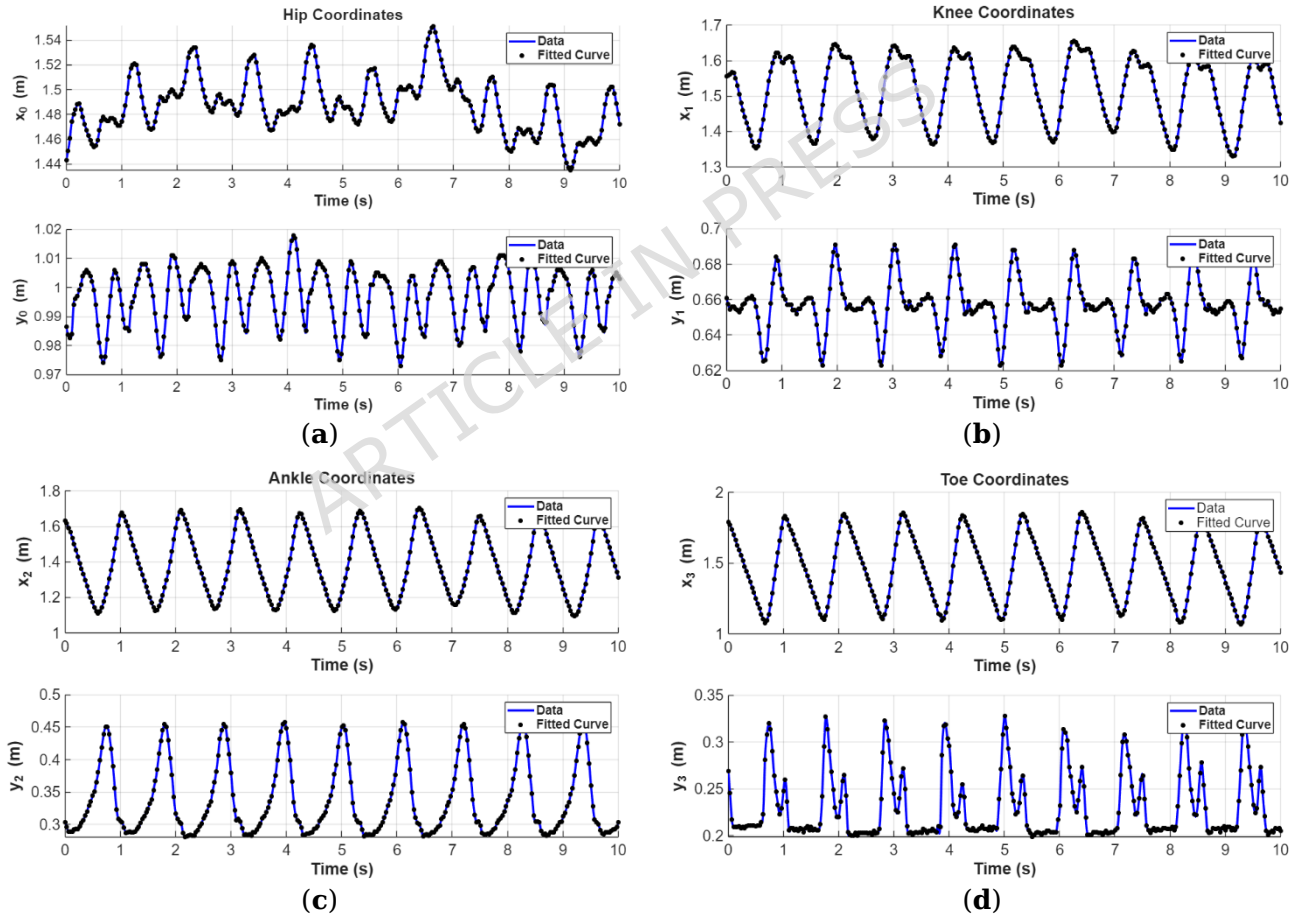
These findings indicate that segment reconstruction accuracy varies across anatomical segments within a 2D planar framework, with distal segments exhibiting greater sensitivity to projection-related effects. Such variability represents an inherent limitation of monocular 2D gait analysis and is expected to influence subsequent inverse-dynamics simulations. Therefore, kinetic results derived from these kinematics are interpreted as model-based trend illustrations rather than precise biomechanical measurements.

**Table 1.** Range and Mean Difference of Segment Lengths Compared to Physical Measurements.

| Segment | Measure (m) | Range from calculation (m) | %Error of Range | Difference between mean and measure (m) | %Error of Difference between mean and measure |
|---------|-------------|----------------------------|-----------------|-----------------------------------------|-----------------------------------------------|
| Thigh   | 0.355       | 0.020                      | 5.634           | 0.003                                   | 0.845                                         |
| Calf    | 0.425       | 0.041                      | 9.647           | 0.058                                   | 13.647                                        |
| Foot    | 0.17        | 0.052                      | 30.588          | 0.007                                   | 4.118                                         |

## Kinematic Calculation and Joint Angle Estimation

Following geometric consistency evaluation of the reconstructed joint coordinates, cubic spline interpolation was applied in MATLAB® to smooth the trajectories, reduce frame-to-frame tracking fluctuations, and generate continuous curves suitable for subsequent kinematic differentiation. The spline fitting was selected to preserve overall motion trends while minimizing high-frequency noise introduced by manual or semi-automatic tracking. Fig. 6 presents the spline-fitted spatial coordinates for all joints.



**Fig. 6.** Fitted spatial coordinate of (a) Hip, (b) Knee, (c) Ankle, (d) Toe.

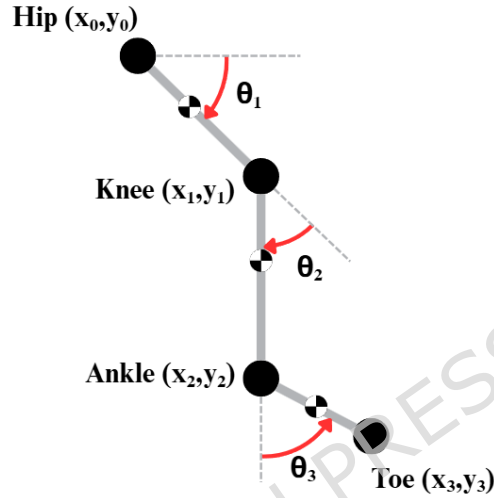
Relative joint angles  $\theta_1$ ,  $\theta_2$ , and  $\theta_3$  were computed from the calibrated sagittal-plane marker coordinates, as illustrated in Fig. 7. These angles describe segment orientations with respect to the horizontal axis and were derived using the geometric relationships defined in Eqs. (2)–(4). The resulting joint-angle profiles are shown in Fig. 8.

$$\theta_1 = \tan^{-1}((y_1 - y_0)/(x_1 - x_0)) \quad (2)$$

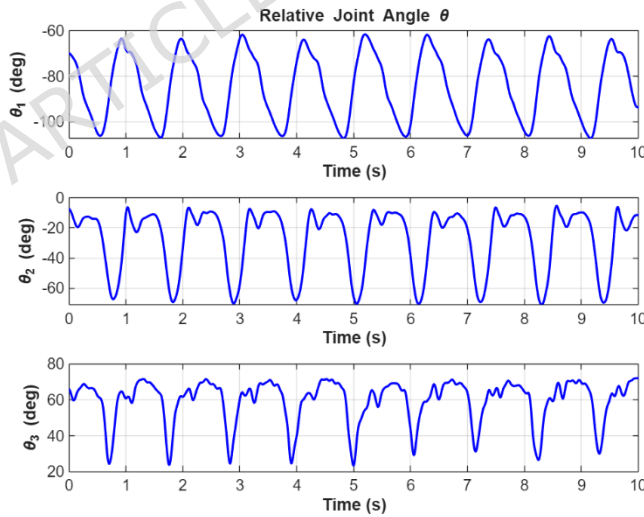
$$\theta_2 = \tan^{-1}((y_2 - y_1)/(x_2 - x_1)) - \theta_1 \quad (3)$$

$$\theta_3 = \tan^{-1}((y_3 - y_2)/(x_3 - x_2)) - \theta_1 - \theta_2 \quad (4)$$

Here,  $\theta_1$  denotes the angle between the horizontal axis and the thigh segment;  $\theta_2$  represents the relative angle between the thigh and calf; and  $\theta_3$  corresponds to the relative angle between the calf and foot. The resulting joint angle profiles are presented in Fig. 8.



**Fig. 7.** Linkage model illustrating human lower limb segments.



**Fig. 8.** Relative joint angle curves.

The corresponding spatial and angular velocities were obtained using a finite-difference method defined as

$$V_{x_i} = \frac{x_{i+1} - x_{i-1}}{2\Delta t} \quad (5)$$

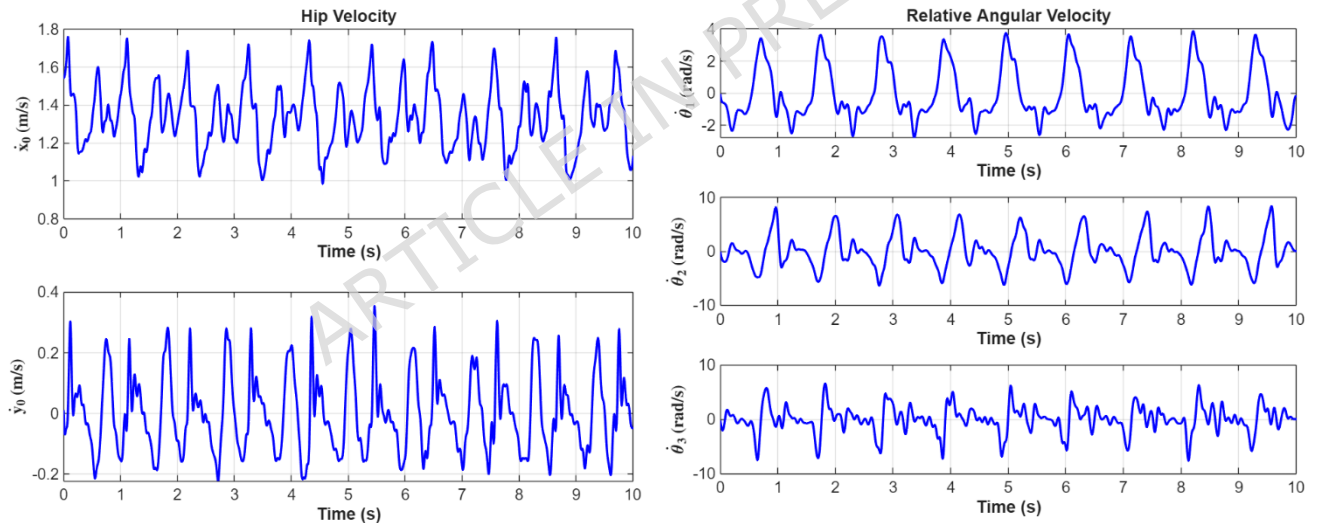
where  $V_{x_i}$  denotes the spatial or angular velocity,  $x_i$  represents the spatial or angular coordinate at sample  $i$ , and  $\Delta t$  is the sampling interval between adjacent measurements  $x_{i+1}$  and  $x_i$ .

To obtain spatial and angular velocities, a central finite-difference scheme (Eq. 5) was employed. Because numerical differentiation can amplify high-frequency tracking noise, the prior spline smoothing step was introduced to mitigate derivative sensitivity. Example hip spatial and angular velocity profiles are presented in Figs. 9(a) and 9(b).

Spatial and angular accelerations were computed using a second-order finite-difference approximation (Eq. 6). As second derivatives are particularly sensitive to small coordinate fluctuations, the resulting acceleration signals are interpreted primarily in terms of trend-level behavior rather than instantaneous peak magnitudes.

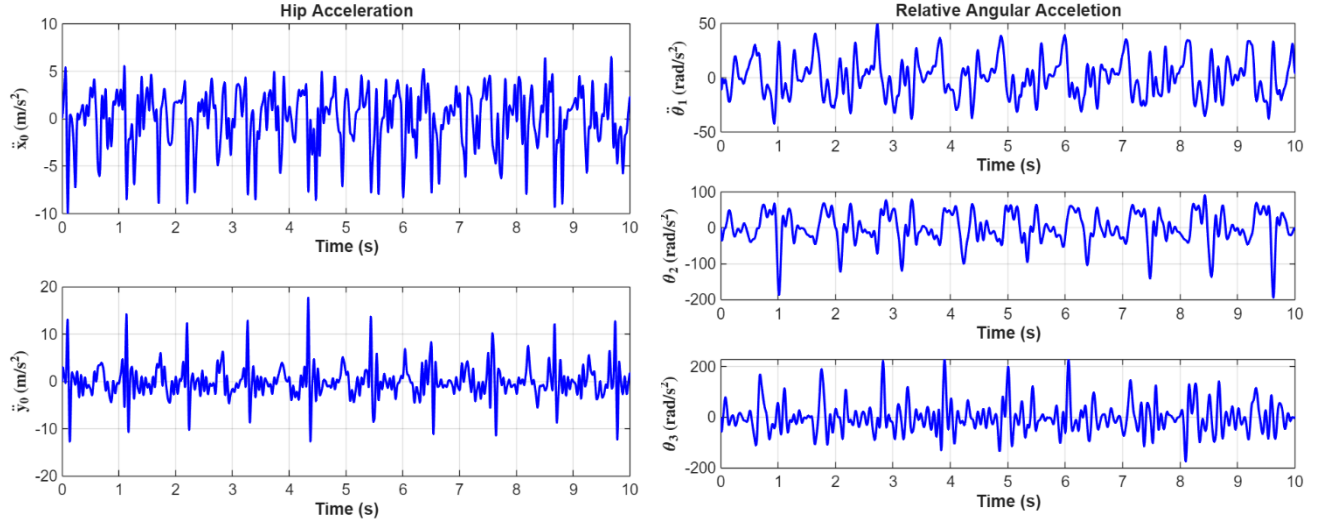
$$A_{x_i} = \frac{x_{i+1} - 2x_i + x_{i-1}}{\Delta t^2} \quad (6)$$

where  $A_{x_i}$  denotes the spatial or angular acceleration. The computed acceleration profiles at the hip are presented in Figs. 9(c) and 9(d), respectively. ( $\ddot{x}$  and  $\ddot{\theta}$  represent the second derivatives of the spatial and angular coordinates, respectively.)



(a) Relationships between hip spatial velocities and time.

(b) Relationships between joint angular velocities and time.



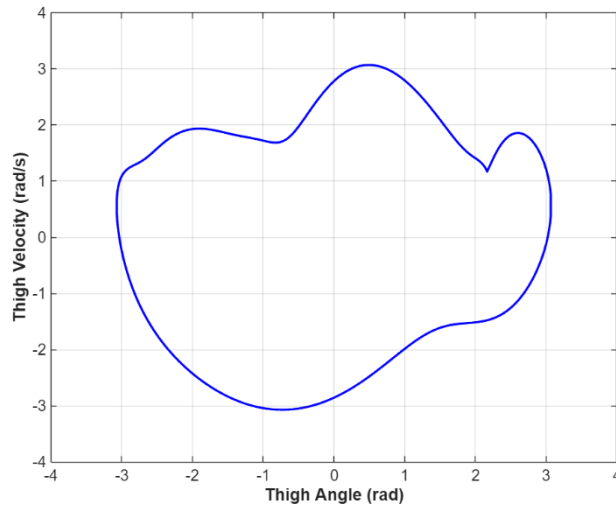
(c) Relationships between hip spatial acceleration and time.

(d) Relationships between joint angular acceleration and time.

**Fig. 9.** Kinematic velocity and acceleration profiles of the hip and lower-limb joints over time.

Due to the periodic nature of walking, all kinematic variables were normalized to gait-cycle percentage rather than absolute time to enable consistent comparison within the analyzed trial.

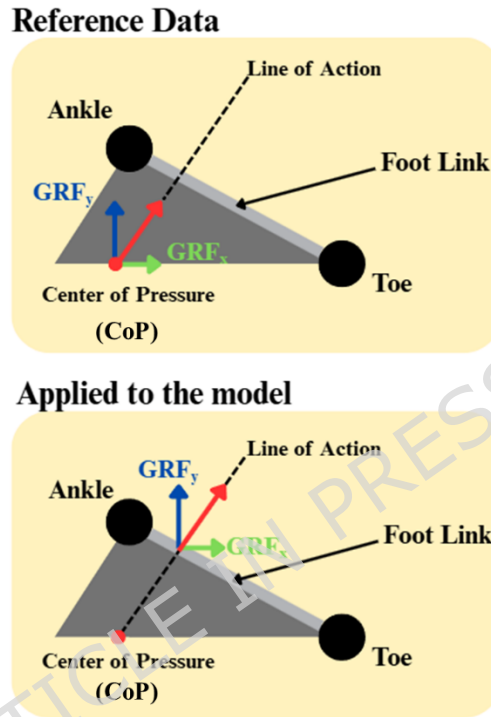
As an illustrative application example, hip joint angle and angular velocity were plotted in the angle-velocity plane to form an Angle-Speed Plot (ASP) representation [29]. The resulting phase diagram (Fig. 10) exhibits qualitative structural similarity to previously reported ASP patterns. This comparison is presented solely to demonstrate the coherence of extracted kinematic trends within the 2D framework rather than to establish clinical or device-level performance.



**Fig. 10.** Normalized Hip angle-speed phase diagram.

## Inverse Dynamics for Joint Torque and Power Illustration

Direct measurement of joint torque and power requires synchronized force-plate measurements and full three-dimensional motion capture. In this study, inverse-dynamics simulations were implemented in MATLAB® Simscape Multibody™ to investigate model-based kinetic behavior using the extracted sagittal-plane kinematics. The lower limb was represented as a planar three-link rigid-body system (Fig. 7), with the hip modeled as a planar joint and the knee and ankle as revolute joints.



**Fig. 11.** Translation of GRF to foot link of the model.

Because the pelvis translates forward during treadmill walking, the horizontal hip trajectory was adjusted by subtracting the treadmill speed to isolate joint-relative motion. Only horizontal translation was corrected because treadmill walking primarily introduces forward belt motion. Since subject-specific ground reaction forces (GRFs) were not measured, a GRF profile and center-of-pressure (CoP) trajectory from [15] were scaled according to the participant's anthropometric parameters and applied to the foot segment as a modeling assumption. The GRF was applied along its line of action and translated from the foot-ground interface to the modeled foot link, as shown in Fig. 11.

The ground reaction force (GRF) profile and center-of-pressure (CoP) trajectory were adopted from the reference dataset reported in [15]. Because the reference dataset reports GRF normalized by body weight, the GRF data were rescaled according to the participant's body mass to obtain the corresponding force magnitudes for the simulation. This assumption is commonly used in simplified inverse-dynamics simulations when subject-specific force measurements are unavailable. Under this assumption, the

simulated torques and powers represent model-driven responses to the imposed kinematic inputs rather than individualized physiological joint loading.

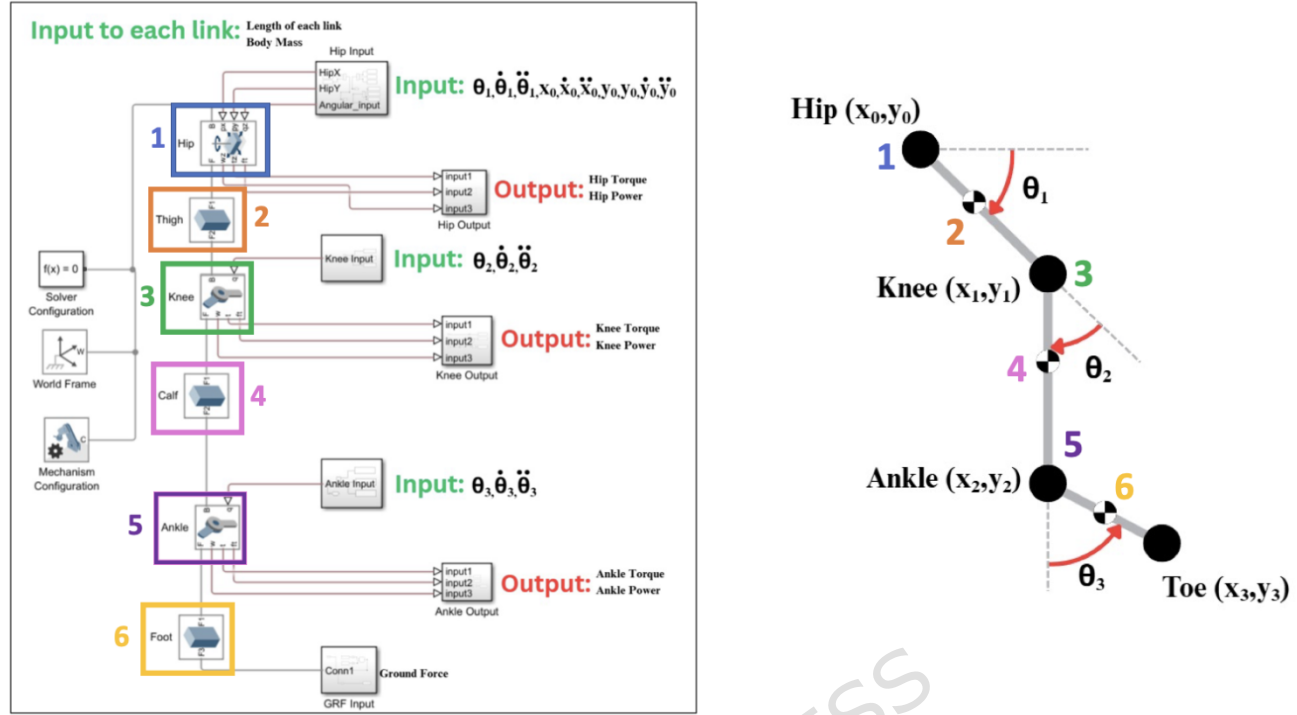
As illustrated in Fig. 12, the lower limb is modeled as a planar three-link rigid-body system consisting of the thigh, shank, and foot segments. The hip joint provides the base motion of the system and its absolute translational position in the global reference frame is described by  $x_0$  and  $y_0$ . The orientations of the three links are defined by the joint angles  $\theta_1$ ,  $\theta_2$ , and  $\theta_3$ , corresponding to the hip, knee, and ankle rotations, respectively.

Consequently, the system kinematics are described by five generalized coordinates ( $x_1$ ,  $y_1, \theta_1, \theta_2, \theta_3$ ). The coordinates of the distal joints ( $x_1, y_1$ ), ( $x_2, y_2$ ) and ( $x_3, y_3$ ) are not independent variables; instead, they are determined geometrically from the segment lengths and joint angles. In the inverse-dynamics simulation,  $x_0$  and  $y_0$  are provided as measured kinematic inputs from the video tracking, while the joint rotations  $\theta_1$ ,  $\theta_2$ , and  $\theta_3$  represent the rotational motion of the three-link mechanism.

The corresponding Simulink implementation used for inverse-dynamics computation is illustrated in Fig. 12. The simulation requires sagittal-plane hip translation and its first- and second-order derivatives; joint angular positions and their derivatives; and segment length, mass, center of mass, and moment of inertia parameters. Segment inertial properties were estimated using anthropometric scaling relationships based on total body mass and segment lengths [15], rather than from subject-specific measurement. Joint torques were computed automatically within Simscape, and mechanical power ( $P$ ) was defined as:

$$P = \tau \dot{\theta} \quad (7)$$

where  $\tau$  is the joint torque and  $\dot{\theta}$  is the relative angular velocity.

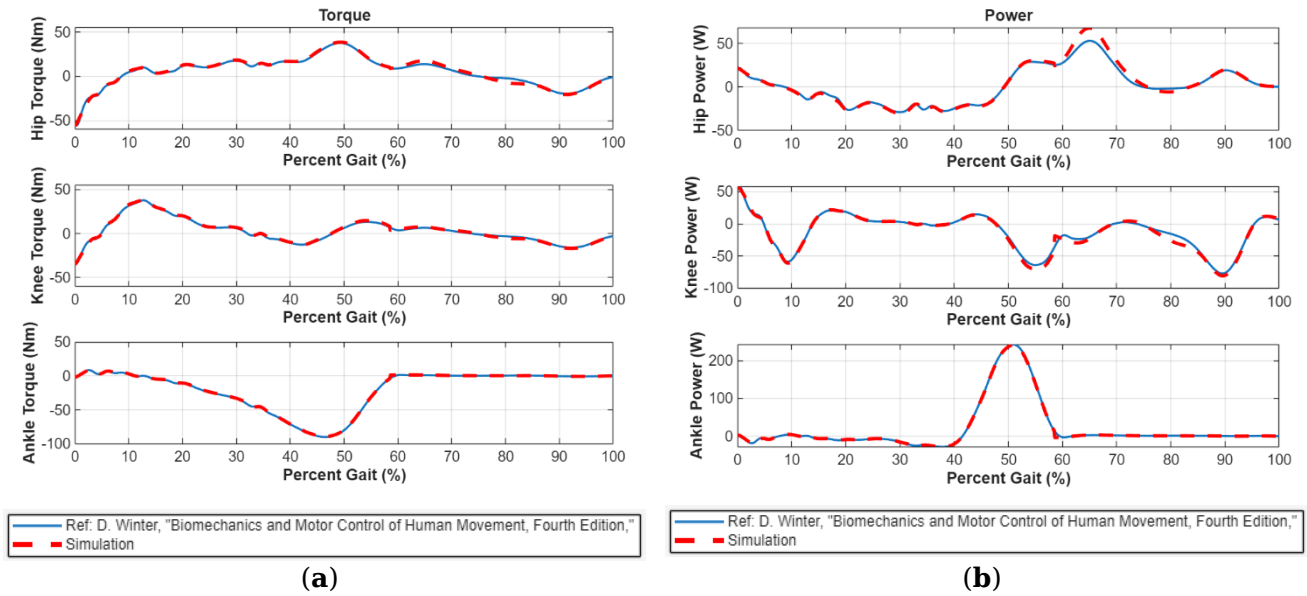


**Fig. 12.** Mapping between block diagram from Simscape and actual three-linkage model.

### Reference-Implementation Consistency Check

To verify internal consistency of the Simscape implementation, the reference kinematic and force data from [15] were first used as simulation inputs. The resulting simulated torques and powers (red dashed curves) were compared with the corresponding reference values (blue curves), as shown in Fig. 13.

Agreement was quantified using normalized root-mean-square (RMS) differences relative to the data range. The normalized RMS differences for torque were 1.8137%, 1.3277%, and 0.5227% at the hip, knee, and ankle, respectively. For power, the normalized RMS differences were 4.7744%, 2.7218%, and 0.4479%, respectively. These results indicate consistent reproduction of the reference dataset within the same modeling framework and confirm correct implementation of the inverse-dynamics formulation in Simscape. This step verifies computational consistency rather than providing independent biomechanical validation of the 2D measurement pipeline.



**Fig. 13. (a) Torque and (b) Power of each joint using reference data.**

### Participant-Based Simulation

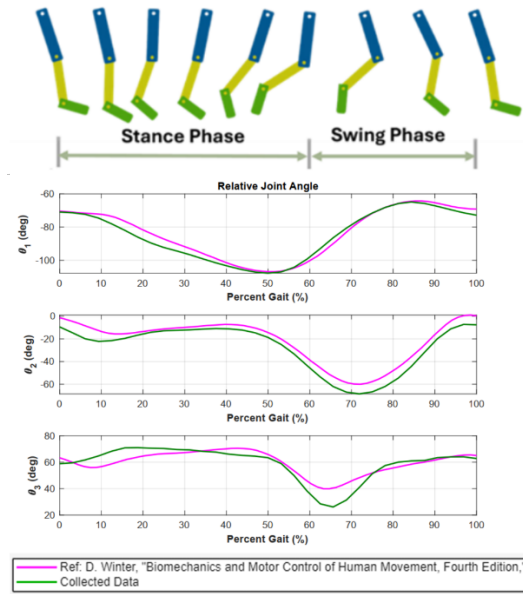
The processed joint-angle data obtained from the participant (Fig. 14) were then applied to the simplified three-link model. The resulting torque and power profiles over one gait cycle are presented in Fig. 15.

Under the imposed kinematic inputs and scaled GRF assumptions, the model-generated peak joint torques were 72.97 Nm at the hip, 92.77 Nm at the knee, and 97.20 Nm at the ankle. The corresponding average absolute joint powers were 35.05 W, 31.22 W, and 21.68 W, respectively. During one gait cycle, negative mechanical work accounted for 71.82% of total joint work at the hip, 61.26% at the knee, and 3.36% at the ankle.

These magnitudes reflect the imposed kinematic trajectories, anthropometric scaling, and reference-based GRF assumptions within the simplified planar model. They are reported to characterize the dynamic behavior of the implemented model and waveform structure across the gait cycle rather than to represent subject-specific physiological joint loading values.

Fig. 16 compares the simulated angular velocity and torque profiles derived from the participant data with those obtained from the reference dataset [15]. Overall, the waveform structures exhibit similar qualitative trends, although differences in magnitude—particularly during early stance—are observed. These deviations likely arise from differences in anthropometry, kinematic inputs, and the use of scaled rather than measured GRF data.

It should be noted that the reference muscle torque and power data from [15] serve as an approximate benchmark for qualitative comparison. Individual variations in segment geometry, mass distribution, and ground reaction forces naturally influence the resulting joint torques and powers.

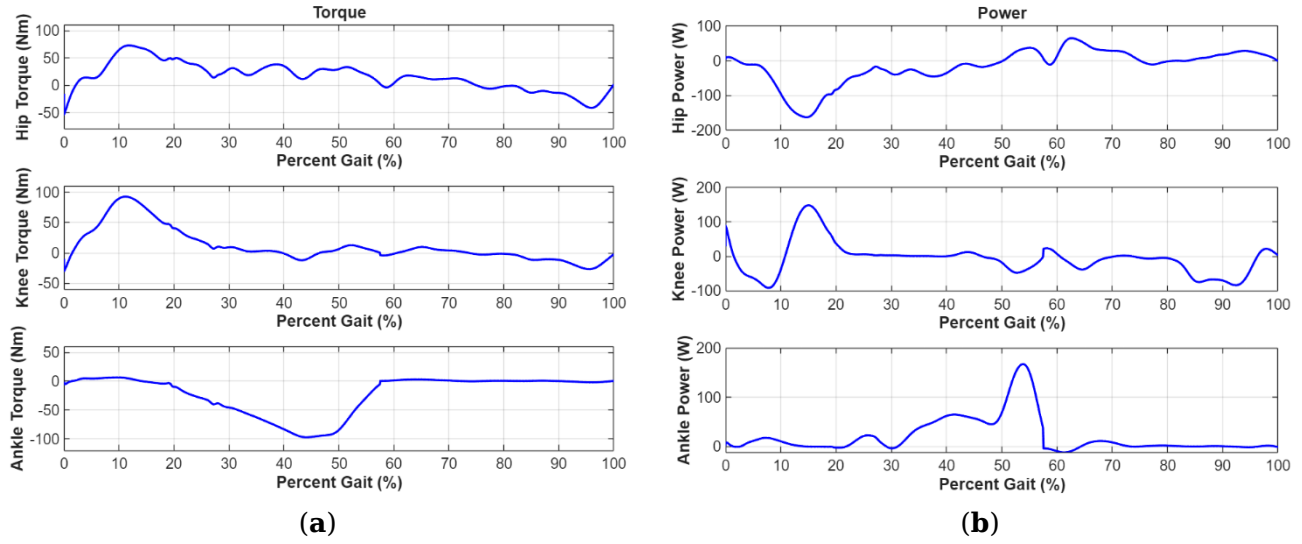


**Fig. 14.** Relative angle of each joint during walking motion in Simscape Multibody.

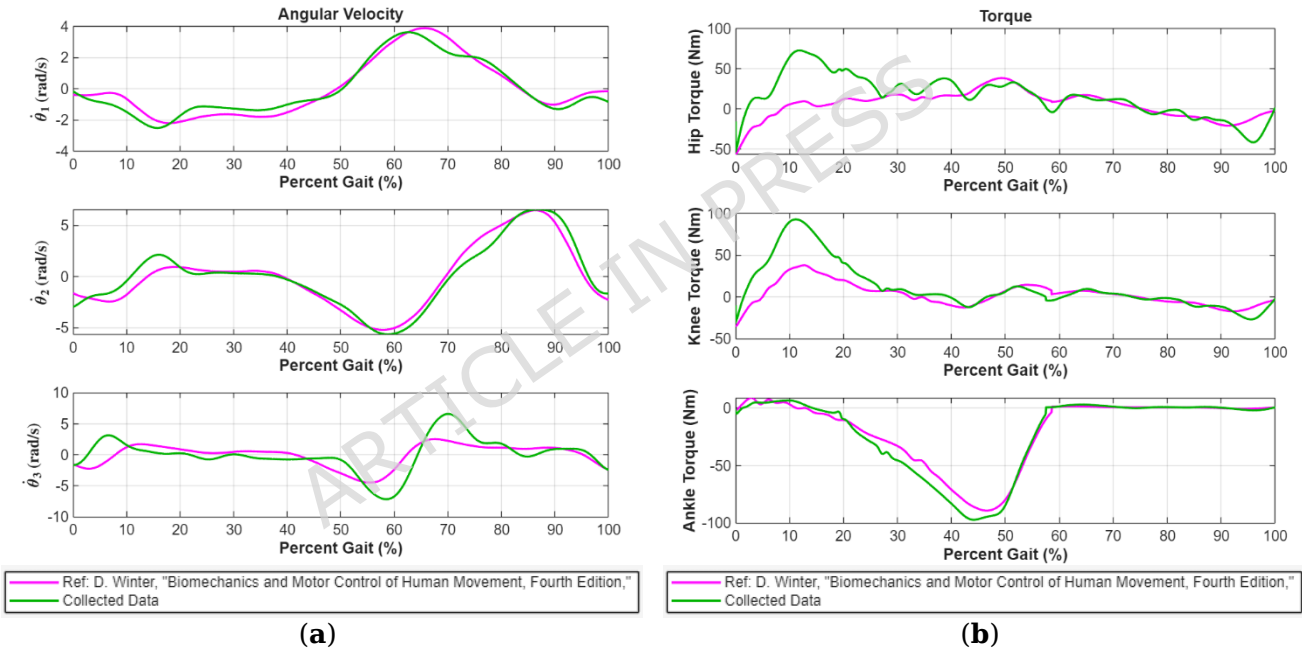
To evaluate whether the extracted kinematic trajectories are consistent with established gait patterns, the joint-angle profiles obtained from the present 2D camera-based measurement were compared with the reference gait dataset reported by Winter [15]. As shown in Fig. 14, the hip, knee, and ankle angle trajectories exhibit similar waveform structures across the gait cycle, particularly in the timing of the major flexion-extension phases and the relative amplitude patterns.

To provide a quantitative assessment of this similarity, the root mean square (RMS) difference and normalized root mean square (NRMSE) were computed between the camera-derived trajectories and the reference dataset for the hip ( $\theta_1$ ), knee ( $\theta_2$ ), and ankle ( $\theta_3$ ) joint angles across the gait cycle. The RMS differences were 3.68 deg., 8.13 deg., and 7.45 deg. and the corresponding normalized RMS values were 0.086, 0.132, and 0.243 for the hip, knee, and ankle joints, respectively. These values indicate that although differences in absolute magnitude exist due to subject variability and simplified modeling assumptions, the overall temporal patterns of the joint-angle trajectories remain consistent with the reference gait dataset.

This comparison suggests that the proposed 2D video-based tracking approach is capable of capturing the characteristic kinematic patterns of human walking in the sagittal plane. It should be noted that the comparison focuses on qualitative waveform structure rather than exact numerical agreement, since the present framework does not include subject-specific anthropometry or three-dimensional motion reconstruction.



**Fig. 15.** (a) Torque and (b) Power of each joint using collected data.



**Fig. 16.** Comparison between simulated results from collected data and from the reference (a) Relative angular velocity (b) Torque.

## Discussion

The Simulink model produced low normalized RMS differences when driven by the reference dataset, indicating consistent reproduction of the published inverse-dynamics results within the same modeling framework. This step verifies the correct implementation of the inverse-dynamics formulation in Simscape prior to applying camera-derived kinematic inputs.

It should be emphasized that the current framework does not aim to provide physiologically validated subject-specific joint kinetics from 2D video input. Instead, the inverse-dynamics component is included as a modeling tool that enables exploratory simulation of torque and power trends based on extracted kinematics under simplified assumptions.

When the participant's extracted kinematics were used as inputs, the resulting torque profiles preserved the characteristic peak-trough waveform structures across the hip, knee, and ankle. However, differences in magnitude—particularly at the hip and knee during early stance—were observed. These discrepancies may arise from variations in data-extraction procedures, individual gait characteristics, walking speed differences, and the use of scaled rather than directly measured ground reaction forces [29].

Joint torques are highly sensitive to GRF magnitude and CoP location. In the present framework, GRF and CoP trajectories were scaled from a reference dataset rather than measured directly, which constitutes a primary modeling assumption. Additional uncertainty is introduced by depth ambiguity inherent to monocular 2D tracking, potential marker occlusion, and overlapping foot regions in the video frames. Because inverse dynamics amplify small kinematic errors, even minor angular inaccuracies or phase shifts may produce noticeable differences in estimated torques and powers. Consequently, the presented torque and power curves should be interpreted as qualitative indicators of gait dynamics rather than quantitative biomechanical measurements. It should also be noted that variability in reconstructed distal segment lengths, particularly in the foot segment, may arise from rapid rotational motion of the foot and projection sensitivity inherent to monocular 2D tracking. Because the toe marker undergoes substantial rotation relative to the ankle during walking, small perspective changes in the sagittal-plane video can produce noticeable frame-to-frame variation in the estimated foot segment length. However, joint angles in the proposed framework are calculated from the relative orientation of adjacent segments rather than from absolute segment-length values. As illustrated in Fig. 14, the reconstructed hip, knee, and ankle angle trajectories remain stable and preserve the characteristic waveform patterns of the gait cycle despite the observed distal tracking variability. Therefore, the distal-segment length variation primarily reflects tracking sensitivity rather than a fundamental limitation of the kinematic reconstruction.

Cubic spline interpolation provided smooth and continuous kinematic trajectories suitable for numerical differentiation. Nevertheless, derivative-based computations remain sensitive to filtering choices and tracking noise. For applications requiring enhanced periodic stability—such as phase-variable estimation—frequency-domain approaches (e.g., Fourier-based fitting) may improve robustness.

Several limitations should be acknowledged. First, the analysis is restricted to sagittal-plane kinematics obtained from a single RGB camera, which cannot capture full three-dimensional motion of the lower limbs. Second, ground reaction force (GRF) inputs were scaled from a reference dataset rather than measured directly for the participant. In conventional gait analysis, GRF measurements and three-dimensional motion capture are typically obtained using instrumented force plates and optical motion-capture systems, which are considered the gold standard for quantifying joint kinematics and kinetics. Third, segment inertial parameters were estimated using anthropometric relationships rather than individualized measurements. Finally, the experimental demonstration was conducted using a single healthy subject under controlled treadmill conditions. The purpose of this participant experiment was not statistical validation but demonstration of the proposed processing pipeline and modeling workflow. Because synchronized ground-truth measurements (e.g., motion capture and force plate data) were not available for the participant experiment, the comparison with the reference dataset was intended only to illustrate qualitative consistency of the reconstructed gait waveform structures. Future work should incorporate subject-specific GRF measurements, multi-plane kinematic capture, and larger participant cohorts to enable rigorous validation of the proposed framework against gold-standard gait analysis systems. Such validation will be essential before the framework can be considered for applications that require physiologically accurate biomechanical measurements, such as clinical gait assessment or assistive-device prescription.

Although the proposed framework does not aim to replace laboratory-grade motion capture systems for precise biomechanical measurement, the extracted kinematic trajectories and trend-level kinetic profiles may still provide useful information for engineering-oriented applications. In particular, joint-angle trajectories and approximate torque patterns obtained from gait analysis are commonly used during the early design stages of wearable assistive devices such as lower-limb exoskeletons. These biomechanical profiles are often required for determining actuator torque requirements, assistance timing, and control strategies in hip- and knee-assistive systems [32]. In addition, hip and knee joint motion characteristics have been widely utilized in biomechanical energy harvesting devices that recover mechanical energy from human walking, particularly during phases of negative joint work [33]. Therefore, while the proposed framework is not intended for clinical diagnosis, it may serve as a practical tool for preliminary gait data acquisition and exploratory biomechanical analysis during the conceptual design and evaluation of wearable assistive technologies.

## Applications

The proposed gait analysis framework is intended for situations where laboratory-based motion capture systems are unavailable or impractical due to cost and infrastructure requirements. By enabling extraction of sagittal-plane joint kinematics from a single RGB camera, the method provides a simple and accessible platform for motion analysis, visualization, and instructional demonstration.

The framework allows reconstruction of joint-angle trajectories and angular velocities during walking, enabling qualitative observation of motion patterns within a simplified planar representation. These kinematic profiles can be used to illustrate temporal relationships between joint variables and to study motion behavior in a controlled modeling environment.

In addition, the extracted trajectories can be used to generate phase-plane representations such as the hip Angle-Speed Plot (ASP), which provides a compact visualization of joint motion dynamics over time. Such representations can serve as conceptual references for understanding cyclic motion behavior and phase transitions in simplified mechanical systems.

The inverse-dynamics component included in this framework generates simulated torque and power profiles based on reconstructed kinematics and simplified modeling assumptions. These results are intended to illustrate general dynamic trends within the model rather than to provide subject-specific biomechanical measurements.

Overall, the framework functions as a low-cost platform for studying planar motion kinematics and basic inverse-dynamics concepts, making it suitable for engineering education, exploratory modeling, and demonstration of motion-analysis principles without requiring specialized laboratory instrumentation.

## Conclusions

This study presents an accessible single-camera framework for estimating sagittal-plane kinematic variables and illustrating model-based kinetic behavior during human gait using a standard RGB camera and open-source tracking software. By integrating 2D video-based joint tracking with an implementation-verified inverse-dynamics model in MATLAB® Simscape Multibody™, the system enables extraction of joint angles and angular velocities, together with simulated torque and power waveforms, without reliance on laboratory-grade motion capture or force-plate instrumentation.

The geometric consistency analysis demonstrated stable reconstruction of segment lengths for proximal segments, while larger variability was observed in distal segments due to projection effects, soft-tissue motion, and the inherent limitations of monocular 2D tracking. The extracted joint-angle and angular-velocity profiles preserved characteristic gait-cycle waveform structures. As expected from planar kinematic input and scaled force assumptions, torque and power magnitudes showed deviations from reference data; however, their temporal patterns remained qualitatively consistent across the gait cycle. These differences arise from modeling assumptions, including reference-based ground

reaction forces, anthropometric scaling, gait variability, and the absence of full three-dimensional motion capture.

The framework is most appropriately interpreted as a tool for trend-level analysis, instructional demonstration, and exploratory research in environments where laboratory systems are unavailable. It is not intended to replace force-plate-based gait laboratories for clinical diagnostics or applications requiring subject-specific joint-loading quantification.

In summary, the proposed 2D gait analysis framework provides a structured and accessible approach for studying sagittal-plane gait kinematics and illustrating inverse-dynamics principles under simplified assumptions. Future work should incorporate subject-specific ground reaction force measurements, multi-view kinematic capture, and enhanced modeling strategies to improve depth representation and quantitative fidelity, including estimation of forces outside the sagittal plane.

## Methods

**Ethics declarations** This study did not require ethical approval by an institutional review board because it involved only a single healthy adult volunteer, with no collection of identifiable personal data, medical interventions, or health-related information. All procedures were non-invasive, and the data collected consisted solely of anonymized motion tracking from video.

**Data availability** Any inquiry can be directly sent to the corresponding author.

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## Ethics approval and informed consent

This study involved a single healthy adult volunteer and did not include any medical intervention, clinical assessment, or collection of personal or health-related data. All procedures were non-invasive and limited to video-based motion capture for engineering and educational purposes.

According to the institutional policies of Chulalongkorn University governing research involving human participants, ethical approval by an institutional review board (IRB) was not required for studies of this nature, which are classified as minimal risk and non-clinical.

All procedures were conducted in accordance with relevant institutional guidelines and regulations, as well as the principles of the Declaration of Helsinki. Written informed consent was obtained from the participant prior to data collection, including consent for the publication of anonymized images in an open-access journal.

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## Author contributions

**Waraporn Phathaisuang** - Conceptualization; Formal analysis; Data curation; Validation; Visualization; Investigation; Writing - original draft; Writing - review & editing.

**Chanoknan Boonlupyanan** - Formal analysis; Investigation; Methodology; Software; Validation; Visualization; Writing - review & editing.

**Parid Vetchaiyo** - Formal analysis; Investigation; Methodology; Software; Validation; Visualization; Writing - review & editing.

**Phatthranit Laocharoen** - Formal analysis; Investigation; Methodology; Software; Validation.

**Peeraphan Limritnonthiphak** - Formal analysis; Investigation; Methodology; Software; Validation.

**Parineak Romtrairat** - Conceptualization; Formal analysis; Methodology; Supervision; Visualization; Investigation; Writing - original draft; Writing - review & editing.

**Thitima Jintanawan** - Conceptualization; Formal analysis; Methodology; Project administration; Supervision; Resources; Visualization; Investigation; Writing - original draft; Writing - review & editing.

**Gridsada Phanomchoeng** - Conceptualization; Formal analysis; Methodology; Project administration; Funding acquisition; Supervision; Resources; Visualization; Investigation; Writing - original draft; Writing - review & editing.

## Competing interests

The authors declare no competing interests.